# Robo-Root

## A Pneumatic Tip-GrowingRobot

Pneumaticeversion,four-tendonsteeringandcamera-based detection in one modular laboratory prototype.

# Rigid robots struggle in confined spaces

## Wheels, tracks and legs

Restricted in narrow passages and complex geometries.

## Tip-growing robots

Extendby eversion: airpressure pushes new tube out of the tip.

## Growth alone gives no directional control.

Theprojecttargetscontrolleddeploymentanddirectionalmovement,with loose, dry granular material in mind.

# The challenge: grow, steer and see together

GROWTH

## Directional control during growth

Airpressure evertsthetube fromitstip,sosteering must be set while the body grows.

TENDONS

## Friction, tension and routing

Four axial tendonsaredrivenbybase-mountedservos and must be routed along the tube.

PERCEPTION

## Reliable perception ahead of the tip

A tip camera with a pretrained YOLO model; 2D boxes do not give obstacle distance.

INTEGRATION

## Linking path planning to hardware through ROS 2

ROS2separatesperception,planning,steering and growth into modules.

# Our solution: four tendons, one camera, ROS 2

## Grow

Pneumaticeversion drives forward growth of the flexible tube.

Growth distance and deployment time are still to be measured.

## Steer

Four servo-driven tendons steer the tip in the X–Y plane. Tendon displacement to tip direction still needs calibration.

## See

Tipcamera plus pretrained YOLO detects objects. Detections give class labels, confidence scores and bounding boxes.

## ROS 2 links perception, planning, steering and growth.

Each function is a separate module, so it can be developed and tested before integration.

10 IEEE PAPERS REVIEWED

# What the literature taught us

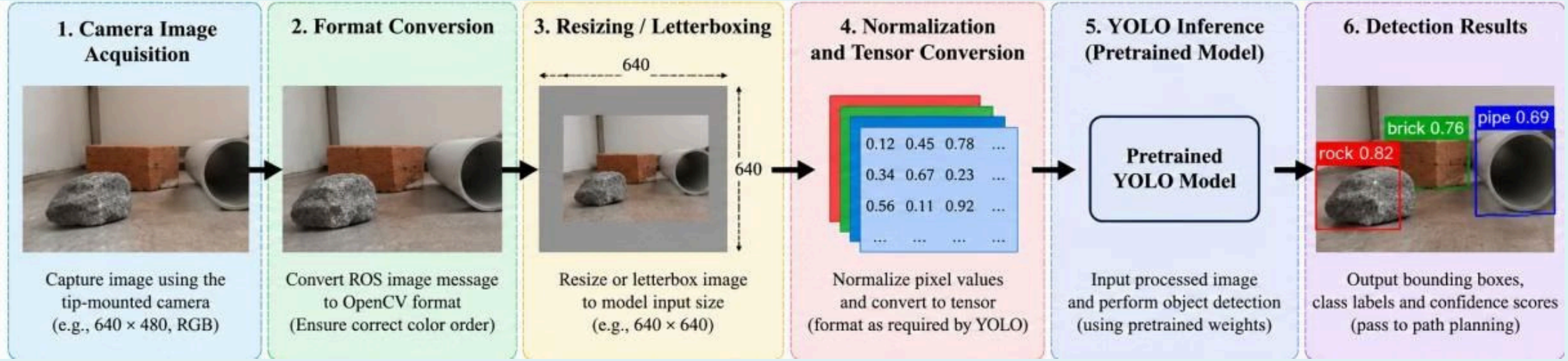
Vine robots, working channels, retraction, sensing, simulation.

Payloads, friction and curved paths affect performance.

- Heap et al.: growth and retraction over long distances, including an industrial pipe route
- Pi et al.: backbone avoided buckling in all 80 straight-length attempts; curved retrieval still deviated
- Kim et al.: self-retraction via a pressurized channel; friction rose with pressure in larger pipes

Our contribution is the integration, not a new growth concept.

# From camera frame to detection

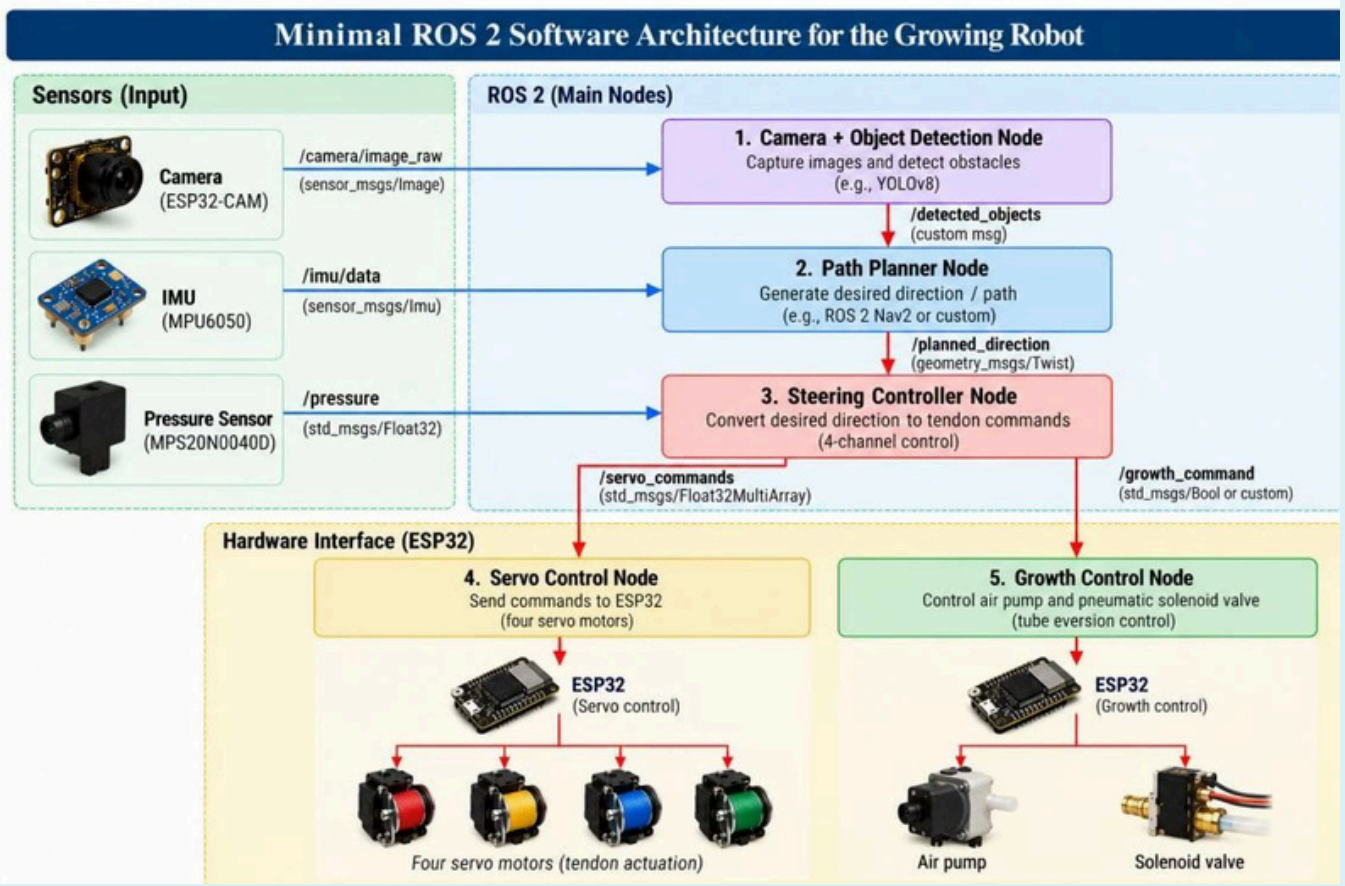



Acquire a tip-camera image (e.g. 640 × 480 RGB), convert ROS to OpenCV format, letterbox and normalize.

Run pretrained YOLO inference, then filter by class, confidence and bounding box.

Detections are forwarded to the ROS 2 path planner; boxes are 2D, so distance needs extra depth sensing or calibration.

# System workflow runs through ROS 2




ROS 2

Camera and detection node, path planner, steering controller, growth controller.

Embedded controller drives servos and pneumatic valve. The path planner turns detections into a desired direction for the steering controller.

# Mechanical design: base station and everting tube

PNEUMATICS

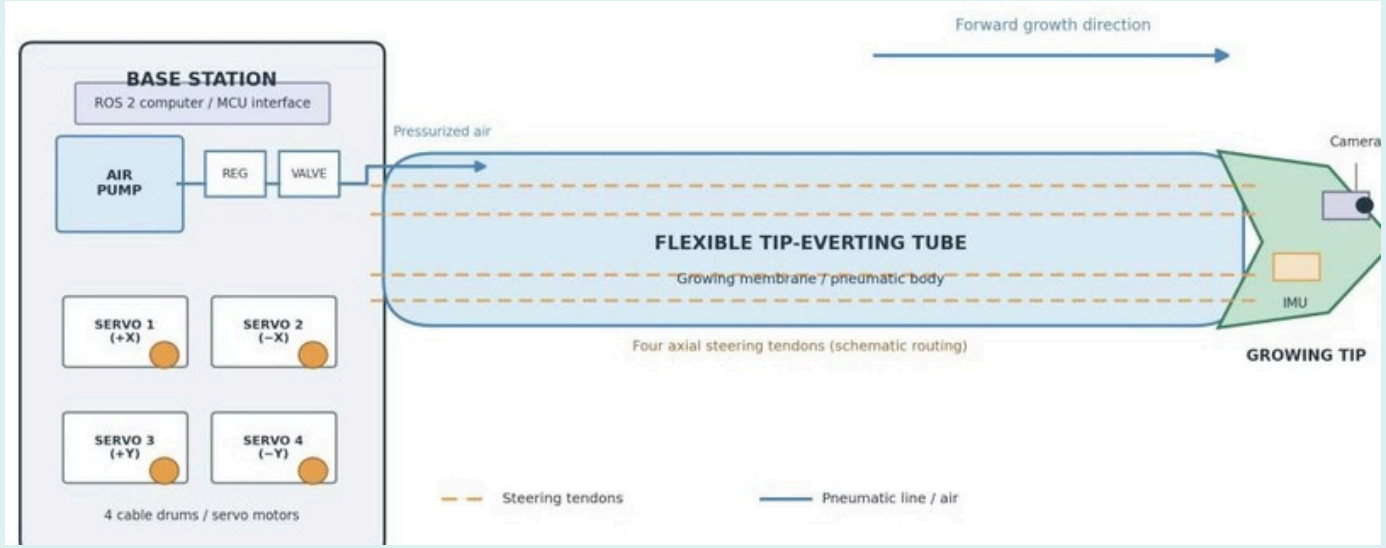
Air pump, regulator and valve feed the flexible tube.

BASE STATION

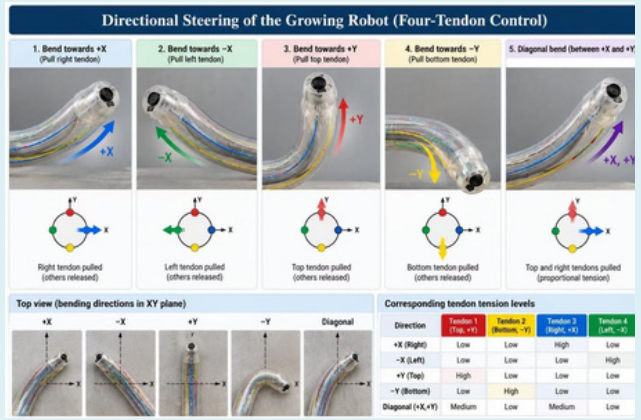
Servos sit at the base, keeping weight off the tip.

GROWING TIP

Camera and IMU sit at the tip; tendons run along the tube.

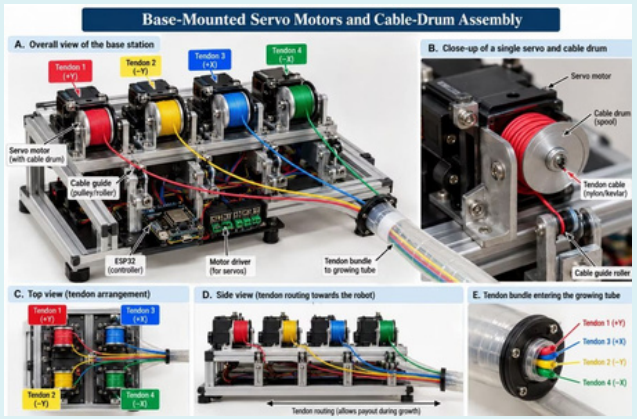


# Steering hardware and electronics



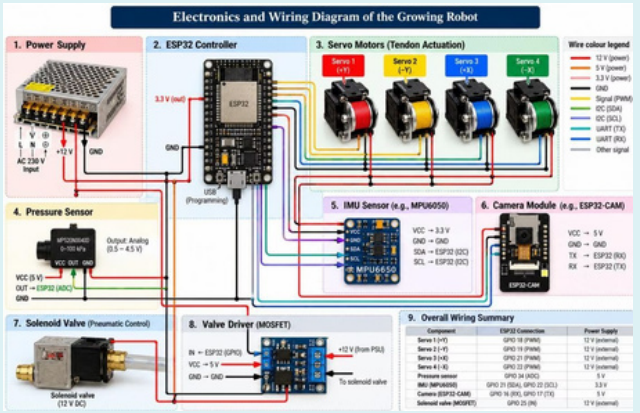

STEERING

Four-tendon control bends the tip: tendons 1–4 pull toward +Y, −Y, +X and −X.




ACTUATION

Servo and cable-drum assembly at the base; four servos pay out cables as the tube grows.




ELECTRONICS

ESP32 controller, valve driver, pressure sensor and IMU; camera module and solenoid valve complete the wiring.

# Conclusion and next steps

## A modular design joining growth, steering and vision

STILL TO VALIDATE

Growth, steering accuracy and detection reliability.

- Growth distance and deployment time
- Steering accuracy and tendon-to-tip response
- Detection reliability of the pretrained YOLO model

NEXT

- Closed-loop steering
- Obstacle-distance estimation
- Deployment and retrieval tests
- Feedback-based steering from camera or IMU data

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